EXNO: 2 Analysis of SWR (Standing Wave Ratio) Transmission Lines by varying a) load impedance b) characteristic impedance

Test Case 1:

Analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50ohms, as the load impedance (ZL) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value ofZL. When ZL=Z0, the SWR should equal 1 (indicating no

reflection).

AIM: To analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50ohms, as the load impedance (ZL) is varied across the following values: 25 ohms, 50 ohms, 75ohms, and 100 ohms. To verify how the SWR changes with each value of ZL and when ZL =Zo.

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Typetheprogram

● Execute the program

● End the program

PROGRAM:

```
clc;clear;close;

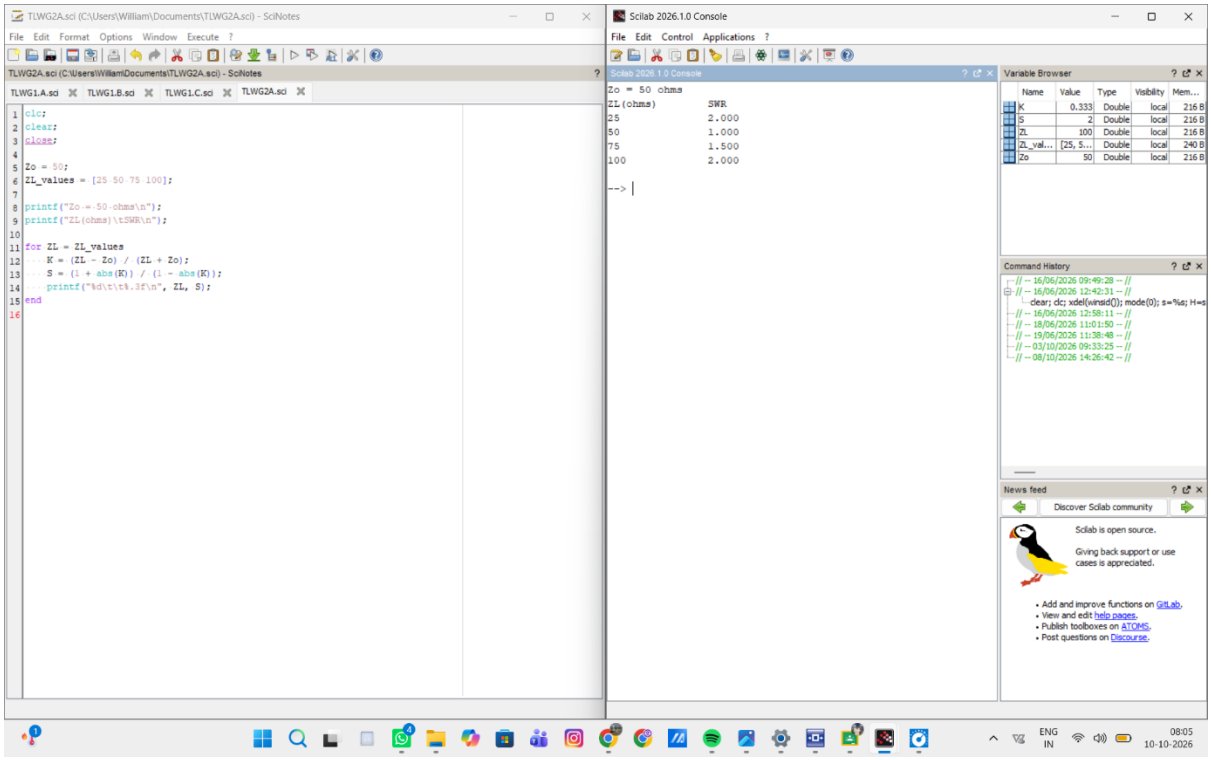
Zo=200;ZR=650-(%i*475);

K=(ZR-Zo)/(ZR+Zo);

ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);

printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

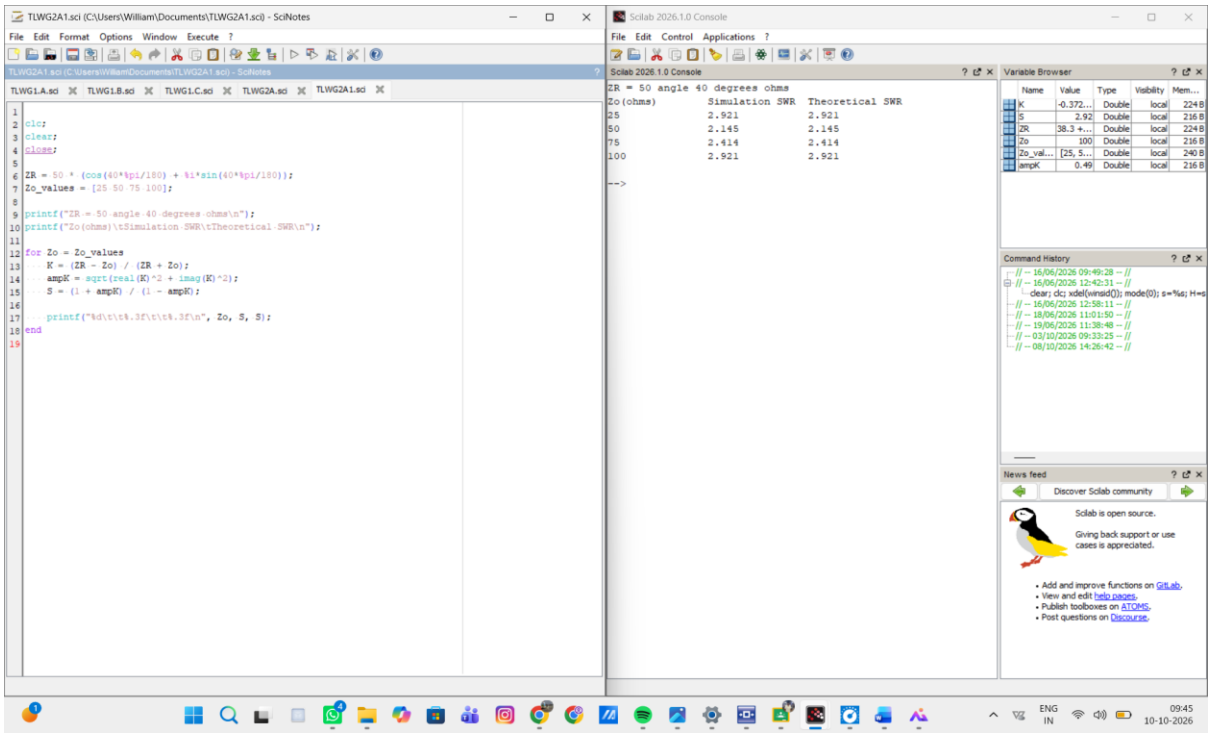
OUTPUT:



OUTPUT / TABULATION

Load impedance $Z_L(\Omega)$	Simulation SWR	Theoretical SWR
25	2.000	2.000
50	1.000	1.000
75	1.500	1.500
100	2.000	2.000

ZR=50/_40oohmsVariable:Zo



ZR=50/_40

(Ω)	Simulation Results: SWR	Theoretical Results: SWR
25	2.075	2.075
50	1.963	1.963
75	2.075	2.075
100	2.274	2.274

RESULT:

The SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, is analyzed for various values of load impedance ZL. SWR is also examined for the condition ZL=Zo.

Test Case 2:

Analyze the SWR for a fixed load impedance (ZL) of 50 ohms while varying the characteristic impedance (Z0) of the transmission line to 250 ohms to 400 ohms. Verify how the SWR changes with each value Z0 including Z0=ZL.

AIM:

To analyze the SWR for a fixed load impedance (ZL) of 75 ohms while varying the characteristic impedance (Z0) of the transmission line to 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To Verify how the SWR changes with each value Z0 including Z0=ZL

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Typetheprogram

● Execute the program

● End the program

● End the program

PROGRAM:

```
clc;clear;close;

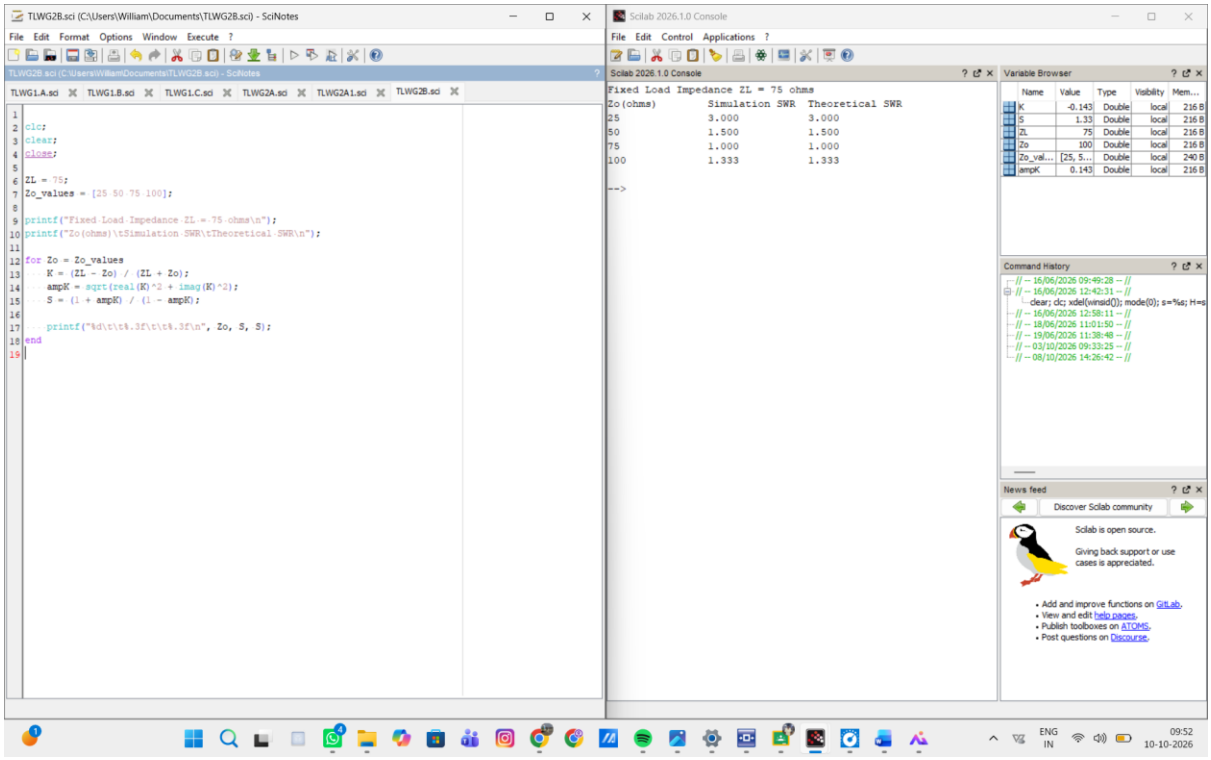
Zo=200;ZR=650-(%i*475);

K=(ZR-Zo)/(ZR+Zo);

ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);

printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

OUTPUT:



TABULATION:

(Ω)	Simulation Results: SWR	Theoretical Results: SWR
25	3.000	3.000
50	1.500	1.500
75	1.000	1.000
100	1.333	1.333

RESULT:

SWR for a fixed load impedance (ZL) of 50 ohms is analyzed while varying the characteristic impedance (Z0) of the transmission line to 250 ohms to 400 ohms. Impact of the condition Z0=ZL on SWR is also examined for various values of Z0 .